25NUEC107 Reg. No. 25141131

ST ALOYSIUS (DEEMED TO BE UNIVERSITY)

Mangaluru

SCHOOL OF ENGINEERING

(UG Programme)

II Internal Exam – November 2025

B. Tech (AIML / ECE) - Semester I

Fundamentals of Electronics

Time: 1 Hour 30 minutes **Max Marks: 50**

Note:
1. This question paper consists of 2 sections. PART – A and PART – B
2. Answer Any 5 questions from Part A.
3. Answer any TWO Full questions from PART B (Q. No. 8 to 11)

PART – A **(2x5=10 Marks)**

1. For a BJT, the ratio of collector current to base current is called _____, and the ratio of collector current to emitter current is called _____.
   a) β, α b) α, β c) V_{BE}, V_{CE} d) I_C, I_B

2. An n-channel MOSFET uses _____ as majority carriers, whereas a p-channel MOSFET uses _____ as majority carriers.
   a) Electrons, Holes b) Holes, Electrons
   c) Electrons, Protons d) Neutrons, Electrons

3. A MOSFET turns ON when _____ exceeds the threshold voltage, resulting in current flow between _____.
   a) V_{GS}, Collector and Emitter b) V_{DS}, Source and Body
   c) V_{GS}, Drain and Source d) V_{DS}, Gate and Drain

4. An enhancement-mode MOSFET _____ have a channel when the gate-source voltage is zero, while a depletion-mode MOSFET _____.
   a) does not, does b) does, does not
   c) does, does d) does not, does not

5. In an ideal Op-Amp, the input impedance is _____, and the output impedance is _____.
   a) Very low, Very high b) Zero, Infinite
   c) Infinite, Zero d) High, High

6. In an ideal Op-Amp, the differential gain is _____, and the common-mode gain is _____.
   a) Infinite, Zero b) Zero, Infinite
   c) High, High d) One, One

7. The input terminals of an ideal Op-Amp draw _____ input current, and the voltage difference between them in a closed-loop configuration is _____.

a) Large, High
b) Zero, Zero
c) Zero, Very small (≈0)
d) High, High

PART – B (2 x 20 =40)

8. a Discuss the Input and output characteristics of a BJT. (10)

b. List the differences between BJT and MOSFET (10)

9. a. Describe the construction and working of an enhancement mode MOSFET. (10)

b. Explain the major currents across the BJT and derive the relationship between α and β (10)

10. a. Enumerate the working of a n-channel JFET along with VI characteristics. (10)

b. Discuss the operation of a depletion type MOSFET. (10)

11. a. Describe any 5 characteristics of an Opamp. Also mention the ideal and typical values of IC 741. (10)

b. **Define Opamps. Explain the internal configurations at the input and output terminals along with the pin configuration of IC 741. Mention any 5 applications of Opamp.** (10)

Mapping of Questions with Bloom's Level and Course Outcomes (COs)															
PART	I							II				III			
Q. No.	1	2	3	4	5	6	7	8a	8b	9a	9b	10a	10b	11a	11b
Blooms Level	2	2	2	2	2	2	2	2	2	2	3	2	2	2	3
Mapped CO(s)	2	2	2	2	3	3	3	2	2	2	2	2	2	3	3

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Date:05.11.2025 **Time:10.00-11.30** **Copies:110**